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ANALYSIS OF VOLUMETRIC LOCKING IN NEARLY INCOMPRESSIBLE HYPERELASTIC MATERIALS USING ABAQUS

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10 June, 2025

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Motivation

- Nearly incompressible materials are widely used in biomedical and industrial applications.
- Standard FEM often fails with these materials due to volumetric locking.
- There's a need for better simulation strategies.



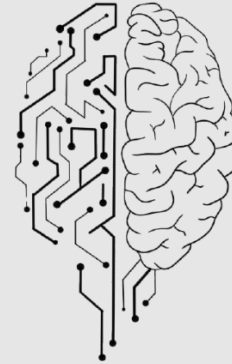
Background

- **Hyperelastic Materials** :Large elastic deformation, recover shape, strain energy-based
- **Nearly Incompressible Materials**: $\nu \approx 0.5$, volume nearly constant, common in soft materials
- **Volumetric Locking**: Artificial stiffness, low displacement, worse with standard elements
- **Standard Elements**: Displacement-only, simple but inaccurate for incompressible cases.
- **Hybrid Elements**: Add pressure field, reduce locking, better accuracy, and more stability.



Objectives

- Analyze volumetric locking in Abaqus
- Compare element types and their effect
- Study mesh type and refinement impact





Key Equations

$$v = \frac{3K - 2\mu}{2(\mu + 3K)}$$

Poisson's ratio

$$W = C_{10}(I_1 - 3)$$

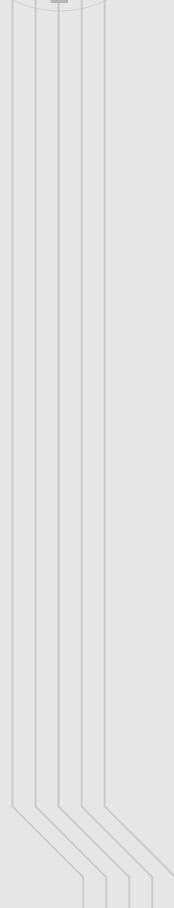
Neo-Hookean

$$K = \frac{2}{D_1}$$

Bulk Modulus

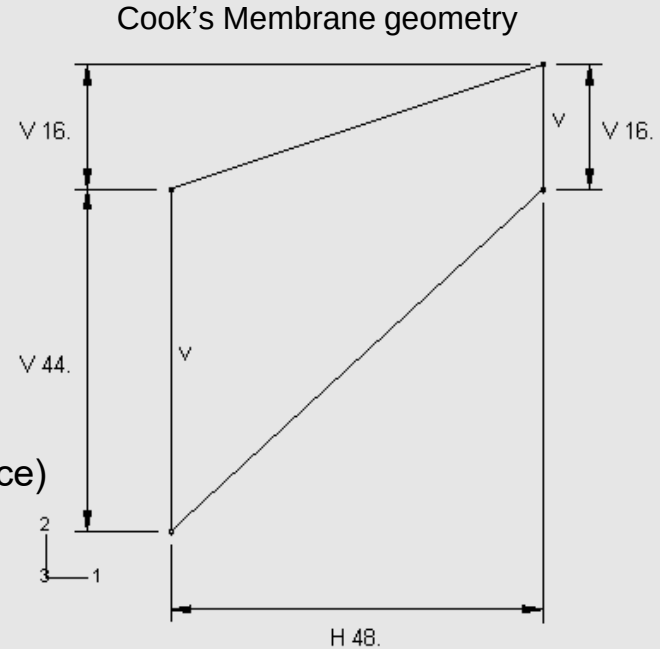
$$\mu_0 = 2(C_{10})$$

Shear Modulus



Methodology

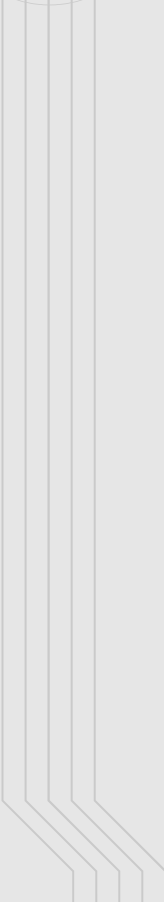
-  **Software:** Abaqus
-  **Material:** Neo-Hookean
-  **Geometry:** Cook's membrane
-  **Loading:** Force & displacement control
-  **Outputs:** U2 (displacement), RF2 (reaction force)



Summary of Element Types Used

Element	Shape	Dim.	Type	Nodes
CPE3H	Triangle	2D	Hybrid	3
CPE4	Quad	2D	Standard	4
CPE4H	Quad	2D	Hybrid	4
CPE4IH	Quad	2D	Hybrid + Incompatible	4
CPE4RH	Quad	2D	Hybrid + Reduced Integration	4
CPE6H	Triangle (2nd)	2D	Hybrid	6
CPE6MH	Triangle (2nd)	2D	Hybrid (Modified)	6
C3D4H	Tetrahedron	3D	Hybrid	4
C3D8H	Hexahedron	3D	Hybrid	8
C3D8IH	Hexahedron	3D	Hybrid + Incompatible	8
C3D8RH	Hexahedron	3D	Hybrid + Reduced Integration	8
C3D10H	Tetrahedron (2nd)	3D	Hybrid	10
C3D10MH	Tetrahedron (2nd)	3D	Hybrid (Modified)	10

Case Studies & Results



Case Study 1: Mesh convergence study for plane strain elements

- **Material constants:**

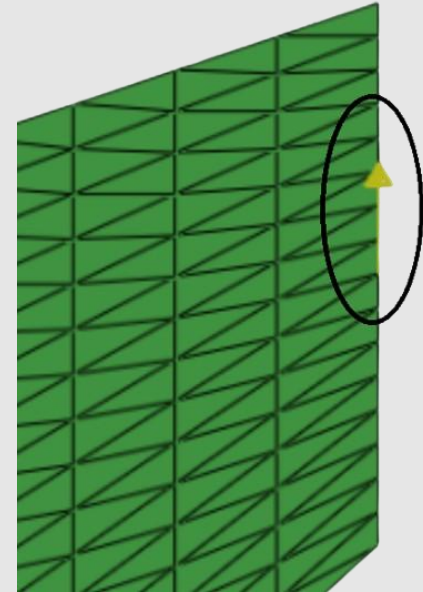
$C10 = 0.4$, $D1 = 2.5e-4$, $K = 8000$ and $\nu = 0.49995$

- **Boundary Conditions:**

Left edge: fixed ($U1 = 0$, $U2 = 0$) and right edge: vertical load applied **1N** (load-controlled)

- **Element types tested:**

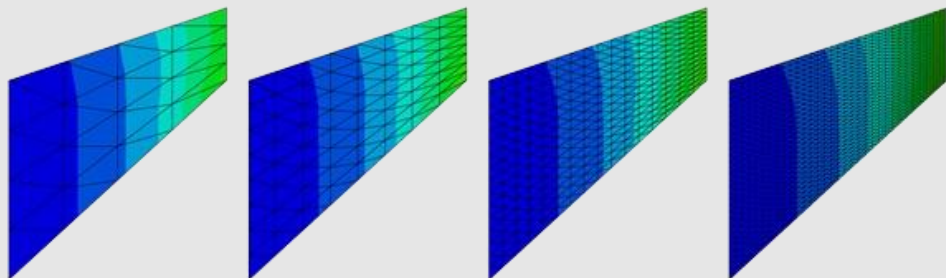
CPE3H, CPE4H, CPE4IH, CPE4RH, CPE6H, CPE6MH



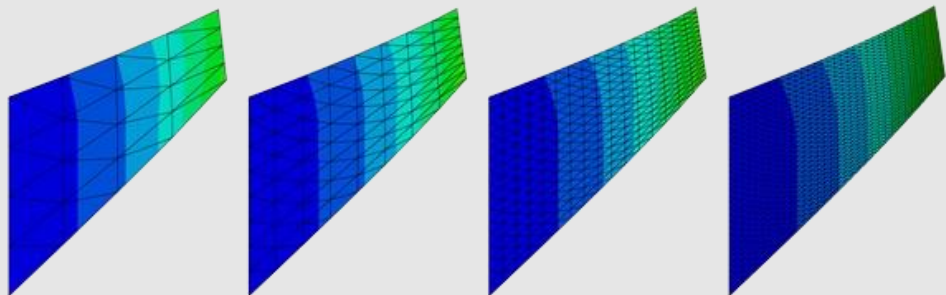
Vertical concentrated load
on the reference point



Case study 1 results

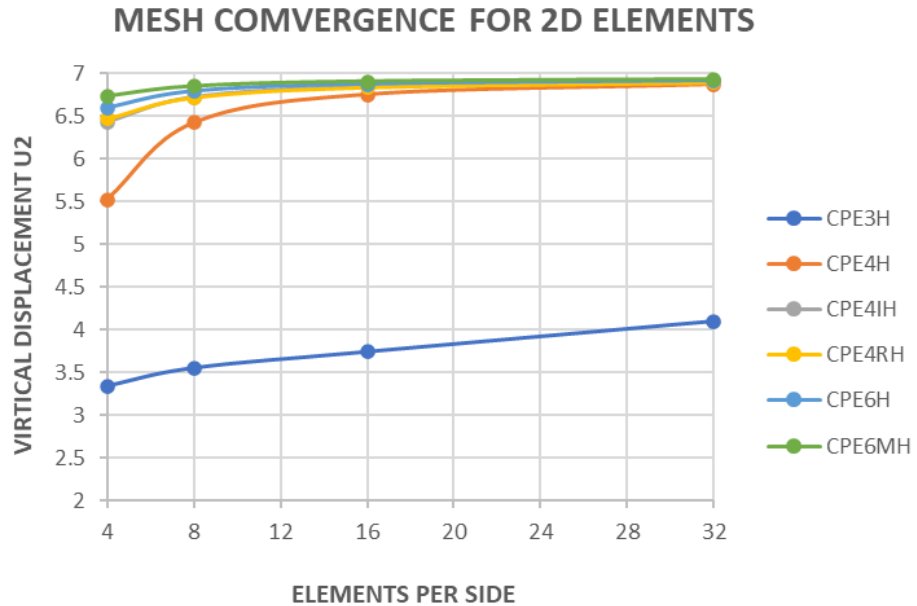


Undeformed



Deformed





CPE3H

- Locking
- Slow convergence

CPE4H / CPE4RH / CPE4IH

- Better accuracy
- Less locking

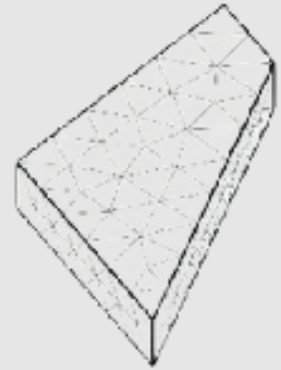
CPE6H / CPE6MH

- Fast convergence
- High stability



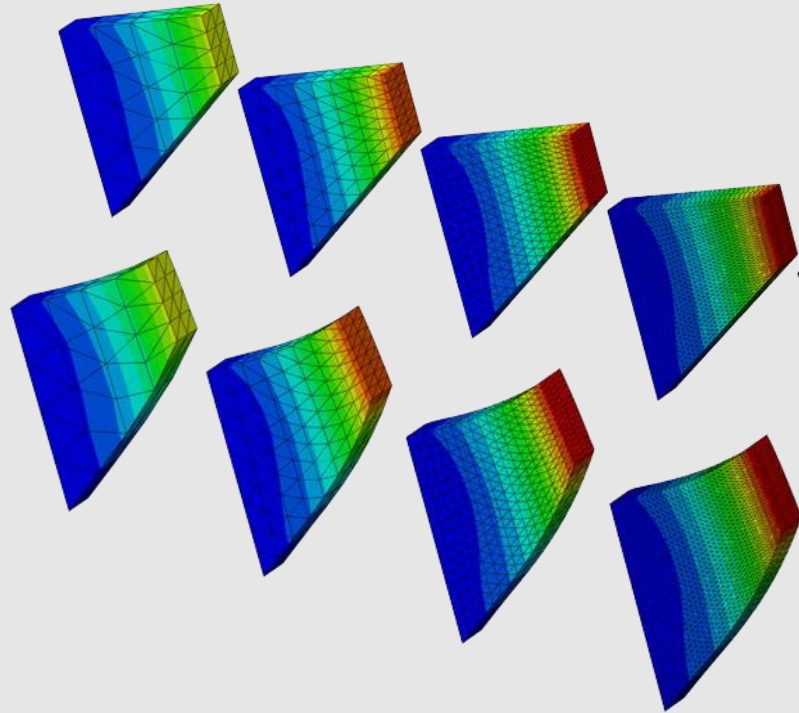
Case Study 2: Mesh convergence study for three-dimensional elements

- **Extend 2D analysis to 3D**
- **Model:** 3D Cook's membrane with 5 mm thickness (out-of-plane)
- **Boundary Conditions:** Symmetry applied on X-Y faces to simulate plane strain behavior
- **Element types tested:** C3D4H, C3D8H, C3D8IH, C3D8RH, C3D10H, C3D10MH





Case study 2 results



Undeformed

Deformed





C3D4H

- Locking
- Poor convergence

C3D8H / C3D8RH / C3D8IH

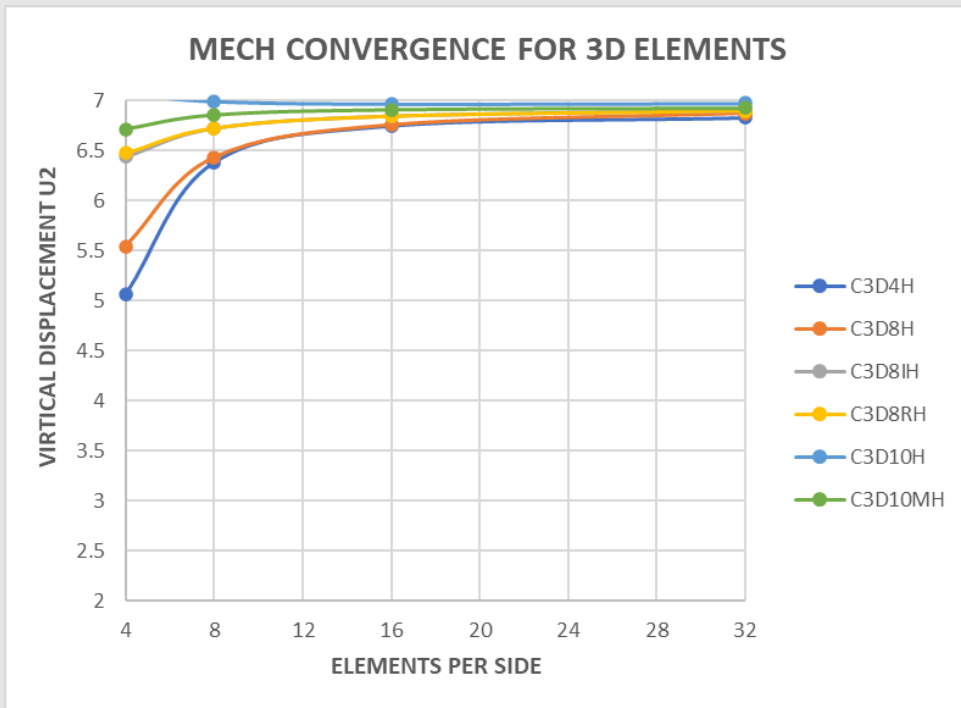
- Better accuracy
- Less locking with mesh refinement

C3D10H / C3D10MH

- Fast convergence
- Most accurate & stable

3D better than 2D:

- Captures full deformation



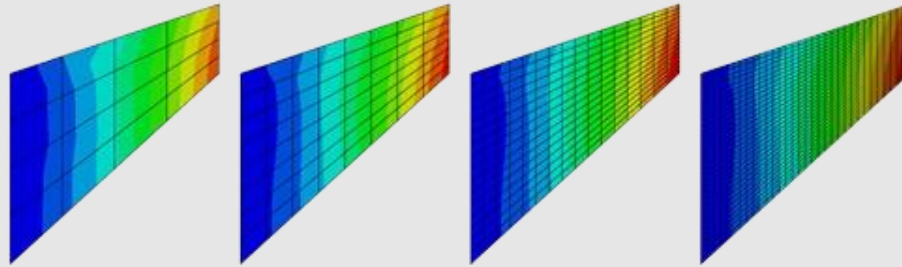


Case study 3: Comparison between CPE4 and CPE4H under varying loading

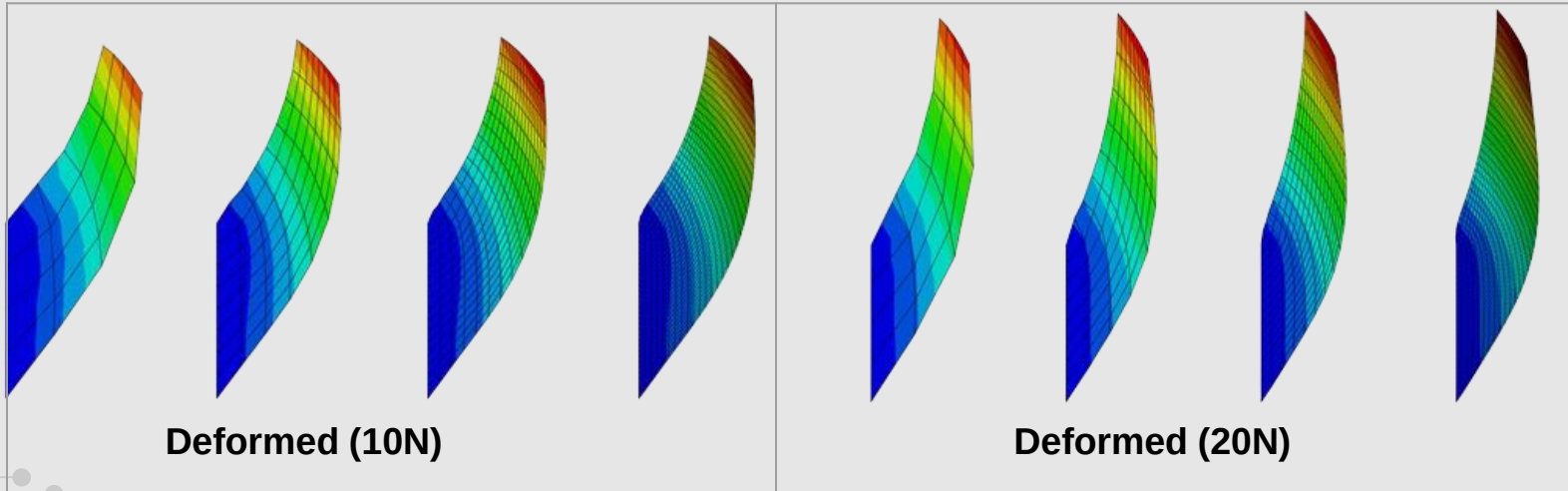
- Objective: Compare standard (**CPE4**) and hybrid (**CPE4H**) elements
- Same Cook's membrane model and boundary conditions
- Loads tested: **10 N** and **20 N**
- Mesh refinement levels: 4, 8, 16, and 32 elements per side
- Goal: Observe volumetric locking behavior and displacement response at higher loads



Case study 3 results



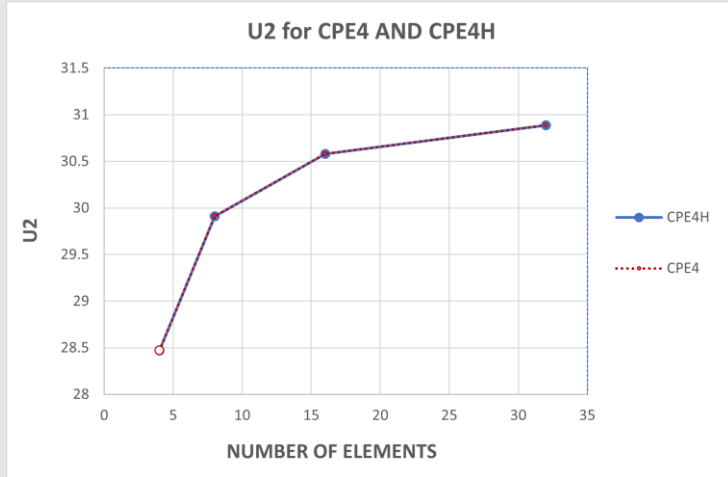
Undeformed



Deformed (10N)

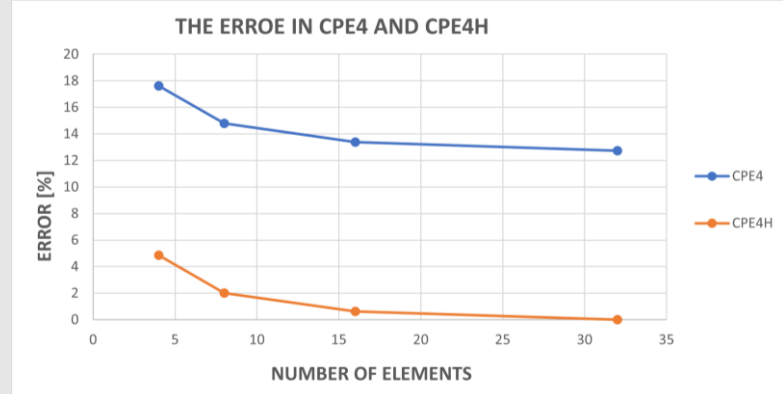
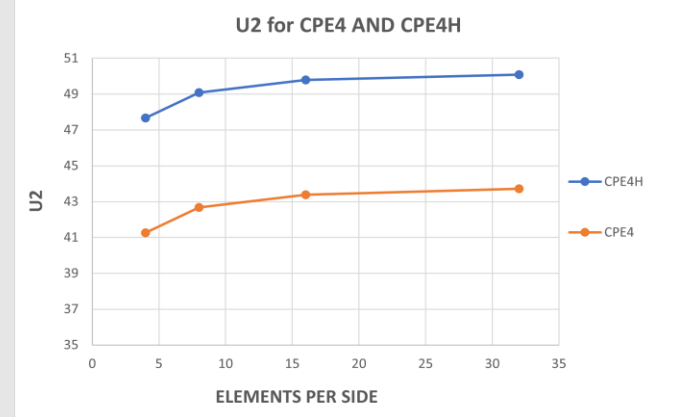
Deformed (20N)

10 N

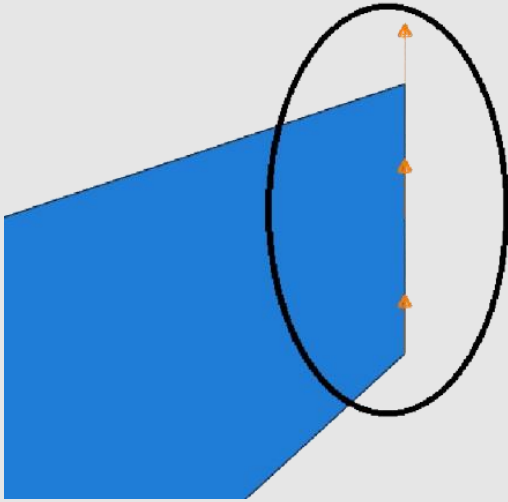


- At 10 N: No clear difference; both elements show similar U2
- At 20 N: CPE4 suffers from volumetric locking; U2 remains low
- CPE4H handles high deformation and converges well

20 N



Case study 4: Reaction force convergence under displacement-controlled



Vertical displacement-controlled (U2)

- **Displacement:**

$U2 = 24 \text{ mm}$

- **Elements:**

CPE4 vs. CPE4H

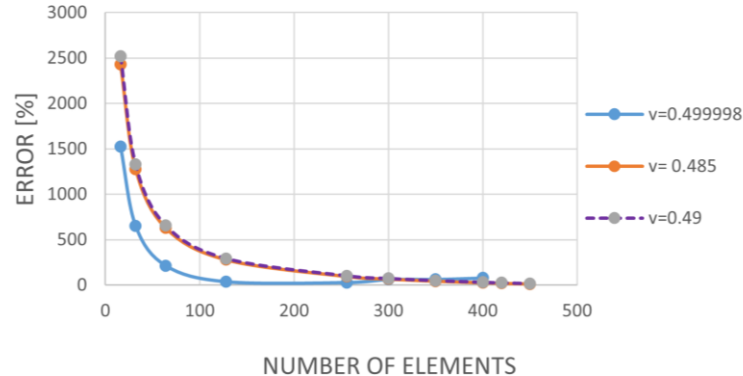
- **Poisson's Ratios:**

0.485, 0.49, 0.499998

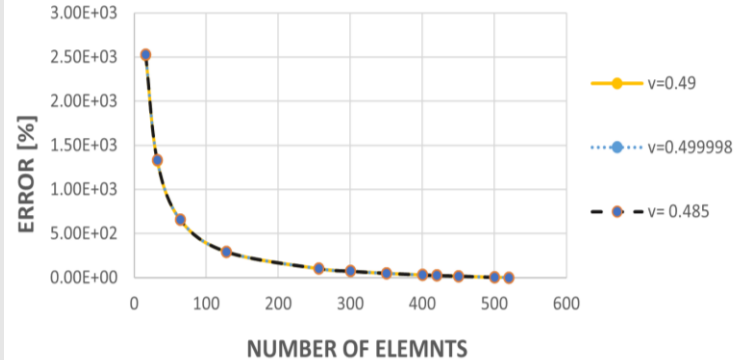


Case study 4 results

Error vs Number of Elements for different Poisson's Ratios in CPE4



Error vs Number of Elements for different Poisson's Ratios in CPE4H

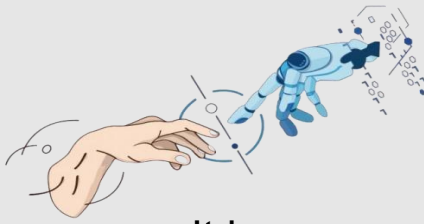


- CPE4 shows high error and slow convergence at $v \approx 0.5$
- Volumetric locking causes instability and inaccurate RF2 results
- CPE4H reduces error significantly across all Poisson's ratios
- The hybrid formulation enables fast and stable convergence
- CPE4H is more suitable for nearly incompressible materials

Conclusion

- Volumetric locking is a major challenge in modeling nearly incompressible materials.
- Standard fully integrated elements suffer from inaccuracy and slow convergence.
- Hybrid elements significantly reduce locking and improve convergence.
- Mesh refinement helps, but is not enough without proper element selection.





Outlook

- It is recommended to select the right element and mesh type to guarantee accurate results.
- It should avoid the standard full integration when analyzing nearly incompressible hyperelastic materials because it leads to volumetric locking.
- The hybrid element is the best type to handle the volumetric locking issue.
- The study can be extended by exploring other hyperelastic material models like Mooney-Rivlin and Ogden to evaluate whether locking behavior is consistent across different constitutive laws





THANK
you

